

BFS

1. Zero path operations

Given a tree with N nodes with some non-negative value assigned to it. Lets say that the value of the i^{th} node is w_i . The beauty of a path between nodes u and v (denoted by $B(u, v)$) is described as the sum of values of all nodes that lie on the path excluding the nodes u and v . Note that $B(u, u)$ is always zero for any valid node u .

Ash is a weird guy. He only likes trees that have $\sum_{u=1}^N \sum_{v=1}^N B(u, v) = 0$. In one operation

Ash can change the value of **any node to any non-negative value**. Ash wants to find the minimum number of operations required so that he starts liking the given tree. But Ash does not have time to solve the problem as he is on his Pokemon adventure. So he needs your help in solving this problem.

Input Format:

The first line contains an integer T denoting the number of test cases. ($1 \leq T \leq 10$)

The following format is followed for each test case:

Next line contains a single integer N denoting the number of nodes in the tree.

($1 \leq N \leq 10^5$)

Next $N - 1$ lines contains two space-separated integers u and v denoting that there is an edge between u and v . ($1 \leq u, v \leq N$)

Next line contains N space-separated integers where the i^{th} integers denotes w_i .

($0 \leq w_i \leq 10^9$)

Output Format:

For each test case, output the minimum number of operations required on a new line.

1	1
3	
1 2	
1 3	
1 2 3	

2. Minimum cost

You are standing at position 1. From position i , you can walk to $i + 1$ or $i - 1$ with cost 1. From position i , you can travel to without any cost to p_i (p is a permutation of numbers $1 \dots n$). You have to reach position n . Determine the minimum possible cost.

Input format

- First line: T denotes the number of test cases ($1 \leq T \leq 10$)
- For each test case:
 - First line: n ($1 \leq n \leq 50\,000$)
 - Second line: n integers where the i_{th} integer is p_i

Note: It is guaranteed that p is a permutation.

Output format

Print the minimum possible cost.

6	3
6	0
1 4 3 2 5 6	0
6	0
3 2 6 5 4 1	1
6	1
3 6 2 4 1 5	
6	
5 4 6 3 2 1	
6	
1 6 5 4 2 3	
6	
5 1 3 6 2 4	

3. Permutation

Given a permutation of 1 to n , you need to perform some operations to make it into increasing order. Each operation is to reverse an interval $a_1, a_2, \dots, a_x (1 \leq x \leq n)$ (a prefix). Your goal is to minimize the number of operations.

Input

The first line contains an integer $n (1 \leq n \leq 8)$.

The second line contains n space separated integers, representing the sequence a .

Output

An integer representing the answer, that is, the minimum number of operations needed to make the permutation into increasing order.

3	2
3 1 2	

4. The Flight Plan

You are given flights route map of a country consisting of N cities and M undirected flight routes. Each city has an airport and each airport can work as layover. The airport will be in two states, Loading and Running. In loading state, luggage is loaded into the planes. In the running state, planes will leave the airport for the next city. All the airports will switch their states from Loading to Running and vice versa after every T minutes. You can cross a city if its airport state is running. Initially, all the airports are in running state. At an airport, if its state is loading, you have to wait for it to switch its state to running. The time taken to travel through any flight route is C minutes. Find the lexicographically smallest path which will take the minimum amount of time (in minutes) required to move from city X to city Y .

It is guaranteed that the given flight route map will be connected. Graph won't contain multiple edges and self loops. A self loop is an edge that connects a vertex to itself.

Input Format:

The first line contains 4 space separated integers, N, M, T and C . Next M lines contains two space separated integers each, U and V denoting that there is a bidirectional road between city U and city V . Next line contains two space separated integers, X and Y .

Output Format:

In the first line print an integer K , denoting the number of city you need to go through to reach city Y from the city X . In next line, print K space separated integers denoting the path which will take the minimum amount of time (in minutes) required by to move from city X to city Y . There can be multiple paths. Print the lexicographically smallest one.

Constraints:

$1 \leq N \leq 10^3$
 $N - 1 \leq M \leq \frac{N*(N-1)}{2}$
 $1 \leq C \leq 10^3$
 $1 \leq T \leq 10^3$
 $1 \leq U, V, X, Y \leq N$

5 5 3 5	3
1 2	1 2 5
1 3	
2 4	
1 4	
2 5	
1 5	

5. Travel diaries

You are given a matrix of size $N \times M$ that contains the digits $0, 1$, or 2 only. All the cells that contain 1 and are adjacent to any cell that contains 2 will be converted from 1 to 2 , simultaneously in 1 second. Write a program to find the minimum time to convert all the cells having value 1 to 2 .

Input format

- First line: Two space-separated integers N and M
- Next N lines: M space-separated integers (denoting the matrix)

Output format

Print the minimum time to convert all the cells having value 1 to 2 .

Constraints

$1 \leq N, M \leq 10^3$

3 5	2
2 1 0 2 1	
1 0 1 2 1	
1 0 0 2 1	

6. Just shortest distance problem

You are given an empty graph of N vertices and M queries of two types:

1. Given a number X answer what is the shortest distance from the vertex 1 to the vertex X .
2. Given two integers X, Y add the new **oriented** edge from X to Y .

Input

The first line contains two integers N and M . The following M lines describe queries in the format mentioned above

Output

For each query of the first type output one integer on a separate line - answer for the corresponding query. Output **-1** if it's impossible to get to the vertex in this query from the vertex **1**.

Constraints

- $1 \leq N \leq 1000$
- $1 \leq M \leq 500\,000$
- $1 \leq X, Y \leq N$

Subtasks

- $N \leq 500, M \leq 5\,000$ in 50 % of test data

4 7	-1
1 4	3
2 1 2	2
2 2 3	
2 3 4	
1 4	
2 2 4	
1 4	

7. Escape from grid

Assume that you are given a two-dimensional grid G that contains N rows and M columns. The grid G consists of only three integers (0, 1, and 2).

- 0 denotes an empty cell
- 1 denotes that a cell contains a plant
- 2 denotes a cell where you are standing initially

You can move into an adjacent cell if that adjacent cell is empty. Two cells are adjacent if they share a side. In other words, if a cell is empty, then you can move in one of the four directions, Up, Down, Left, and Right.

You cannot move out of the grid G .

Your task is to find the length of the shortest path to reach one of the boundary edges of the grid without stepping on a plant. The length of a path is equal to the number of moves you make.

Note: It is guaranteed that there is only one cell with value 2 (you are standing in only one cell initially).

Input format

- First line: Two space-separated integers, N and M , denoting the number of rows and columns in the grid G respectively
- Next N lines: M space-separated integers denoting the rows of the grid G

Output format

Print the length of the shortest path to reach one of the boundary edges of the grid without stepping on a plant. If it is not possible to reach the edge of the grid, then print **-1**.

Constraints

$$1 \leq N, M \leq 100$$

$0 \leq G_{i,j} \leq 2$ where $G_{i,j}$ denotes a cell of the G grid

4 5	2
1 1 1 0 1	
1 0 2 0 1	
0 0 1 0 1	
1 0 1 1 0	

8. Robotic paths

You have built a new robot and placed it in a maze. The maze contains N rows and each row contains M cells. Each cell of the maze is either an **empty cell** or an **obstacle**. An empty cell is denoted by a (.) dot and an obstacle is denoted by a (#) hash. The robot is operated by your computer with some keyboard commands.

Currently, there are 4 character commands:

- L : Moves the robot from the current cell to its **left** cell
- R : Moves the robot from the current cell to its **right** cell
- U : Moves the robot from the current cell to its **upper** cell
- D : Moves the robot from the current cell to its **lower** cell

The robot can go to some cell only if it is empty and inside the maze. Therefore, you cannot give a command that takes the robot to an obstacle or out of the maze.

The robot is currently at (s_x, s_y) cell. You want to take it to (f_x, f_y) cell. Therefore, you are required to build a string of character commands that takes the robot to the destination. There may exist several strings that can perform the same task but you must use the string of the minimum length. If there exist several strings of the minimum length, you must select the lexicographically smallest string of them.

Refer to the sample explanation section for better understanding.

Input format

- First line: Two integers N ($3 \leq N \leq 50$) and M ($3 \leq M \leq 50$)
- Next N lines: Strings of M lengths that represent the maze
- Next line: An integer Q ($1 \leq Q \leq 300$) denoting the number of queries
- Each of the next Q lines: Four integers s_x, s_y, f_x , and f_y

Note: It is guaranteed that (s_x, s_y) and (f_x, f_y) will be empty cells and $(s_x, s_y) \neq (f_x, f_y)$

Output format

If it is possible to reach the destination, then print the string command as mentioned. Otherwise, print *Impossible*.

5 5	LDDR
.....	DRRU
..#..	UUUR
.....	Impossible
#.##.	DDRRRRDD
..#...	
5	
1 3 3 3	
2 2 2 4	
4 2 1 3	
1 1 5 1	
1 1 5 5	

9. Level Nodes

You have been given a Tree consisting of N nodes. A tree is a fully-connected graph consisting of N nodes and $N-1$ edges. The nodes in this tree are indexed from 1 to N . Consider node indexed 1 to be the root node of this tree. The root node lies at level one in the tree. You shall be given the tree and a single integer x . You need to find out the number of nodes lying on level x .

Input Format

The first line consists of a single integer N denoting the number of nodes in the tree. Each of the next $n-1$ lines consists of 2 integers a and b denoting an undirected edge between node a and node b . The next line consists of a single integer x .

Output Format

You need to print a single integers denoting the number of nodes on level x .

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq a, b \leq N$$

Note: It is guaranteed that atleast one node shall lie on level x

20	3
11 1	
1 2	
13 3	
15 4	
17 5	
11 6	
2 7	
1 8	
15 9	
4 10	
15 12	
5 13	
2 14	
17 15	
15 16	
11 17	
15 18	
9 19	
16 20	
2	

10. Prime cells

You are given a matrix M of R rows and C columns. You are required to maximize the size of a prime cell. A prime cell is the largest group of connected cells. A cell may only be connected with its adjacent cells in the direction up, down, left, and right that has the same count of prime factors. You are allowed to perform any number of operations. In one operation, you can change a number to another number. A cost is associated to change the number. This cost is the absolute difference between the highest prime factors of both the numbers. The maximum value of the modified matrix should not exceed the maximum value of the input matrix.

Note: Assume that the highest prime factor and the count of prime factors for 1 is 1. If two prime cells with the same size are found the one with the largest score will be considered.

Score evaluation

Your score is calculated as follows:

$$Score = \frac{(Sum\ of\ elements\ in\ the\ prime\ cell * Size\ of\ the\ prime\ cell)}{(1 + Total\ Cost)}$$

Your task is to maximize the score.

Input format

- First line: Two space-separated integers R and C denoting the number of rows and the number of columns respectively
- Next R lines: C space-separated integers denoting the value of matrix $M[i][j]$

Output format

Your task is to print the two space-separated integers R and C denoting the number of rows and the number of columns respectively. These integers must be followed by the modified matrix from a new line.

Constraints

$$1 \leq R, C \leq 10^3$$

$$2 \leq M[i][j] \leq 10^6$$

```
3 3
2 3 24
21 60 18
7 11 330
```

```
3 3
2 3 24
21 3 18
7 11 330
```

11. Printing patterns

You are required to form a matrix of size $r \times c$ where r is the number of rows and c is the number of columns. You are required to form the waves of numbers around the provided center, (C_i, C_j) (0-based indexing).

Example:

- Size of the matrix: $r = 9$ and $c = 9$
- Center coordinates: $C_i = 4$ and $C_j = 4$ (0 based indexing)
- Pattern:

```
4 4 4 4 4 4 4 4 4
4 3 3 3 3 3 3 3 4
4 3 2 2 2 2 2 3 4
4 3 2 1 1 1 2 3 4
4 3 2 1 0 1 2 3 4
4 3 2 1 1 1 2 3 4
4 3 2 2 2 2 2 3 4
4 3 3 3 3 3 3 3 4
4 4 4 4 4 4 4 4 4
```

You are given the values of r, c, C_i, C_j (the values of C_i and C_j is 0-based indexed). Your task is to print the provided pattern.

Input format

The first line contains four integers r, c, C_i , and C_j denoting the number of rows, number of columns, x coordinate of center, and y coordinate of center.

Output format

Print the pattern for the provided input.

Constraints

- $1 \leq r \leq 1e3$
- $1 \leq c \leq 1e3$
- $0 \leq C_i < r$
- $0 \leq C_j < c$

10 10 5 5

5 5 5 5 5 5 5 5 5 5
5 4 4 4 4 4 4 4 4 4
5 4 3 3 3 3 3 3 3 4
5 4 3 2 2 2 2 2 3 4
5 4 3 2 1 1 1 2 3 4
5 4 3 2 1 0 1 2 3 4
5 4 3 2 1 1 1 2 3 4
5 4 3 2 2 2 2 2 3 4
5 4 3 3 3 3 3 3 3 4
5 4 4 4 4 4 4 4 4 4